



AFTER

Developments in military technology under way at the time of the Civil War would make offensive infantry tactics even more problematic over the next 50 years.

REPEATING RIFLES

The infantry weapons that would make the rifle musket obsolete were **breech-loading rapid-fire guns** using metal cartridges. Repeater rifles, especially the Spencer rifle, were used quite widely in the Union army by the end of the war. **The Spencer could fire 20 rounds a minute**, compared with the maximum three rounds of a muzzle-loader. The **Gatling gun**, used experimentally in the Civil War, was capable of firing 200 rounds a minute and heralded the era of automatic weapons.

TRENCH WARFARE

In the decades after the Civil War, **artillery also became dramatically more efficient** in range, rate of fire, and the explosive power of shells. Along with the use of barbed wire to protect entrenchments, these weapons developments led to the trench warfare of World War I, where the **same fundamental infantry tactics were employed**, with even higher casualty rates for even less result.